

SUPPLEMENTARY INFORMATION

Machine Learning-Assisted Multi-Objective Optimization of Battery Manufacturing from Synthetic Data Generated by Physics-Based Simulations

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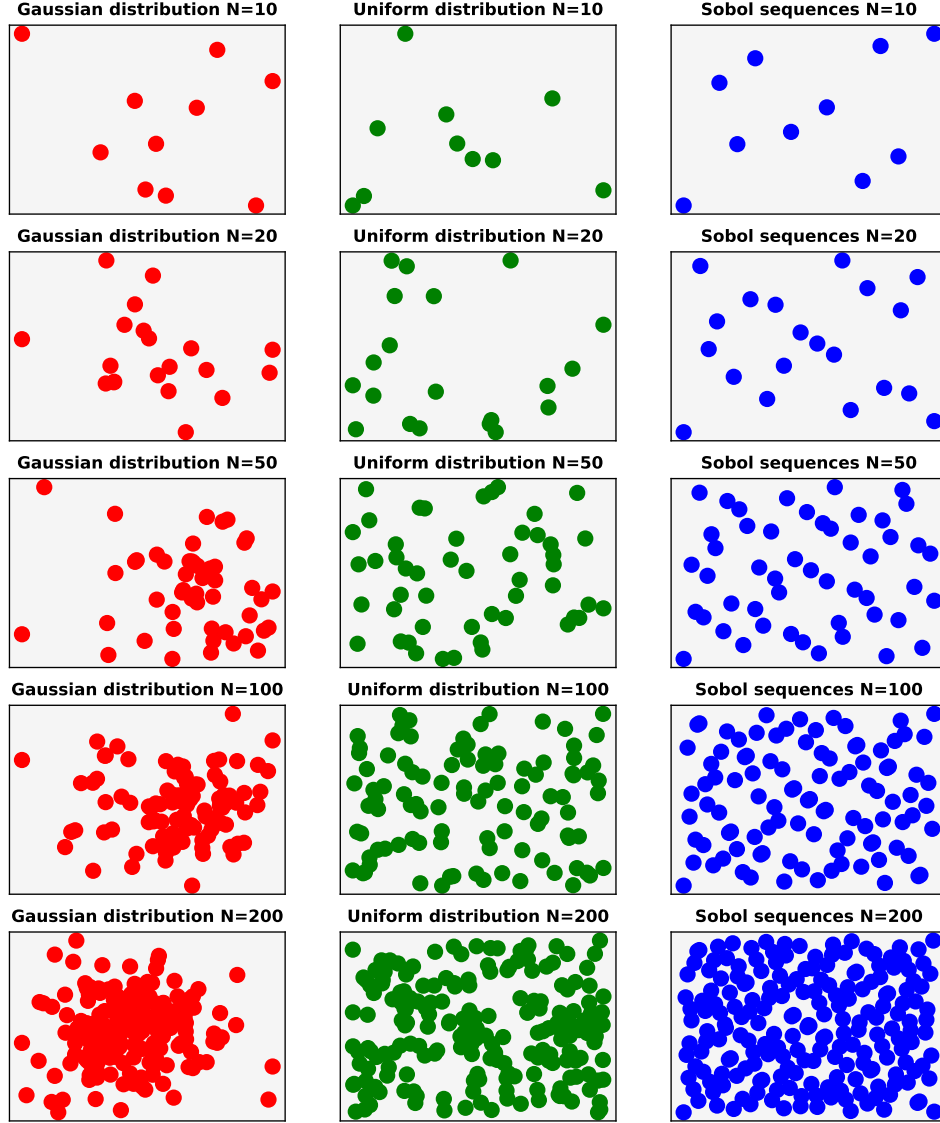


Figure S1 : 2D Graphical representation of Gaussian distributions (red), uniform distributions (green), and Sobol sequences (blue), for different amount of data points in the space $[0, 1]^2$. The comparison highlights that the data points color-coded in blue fill in a more uniform way the 2D space compared to the two other types of distributions.

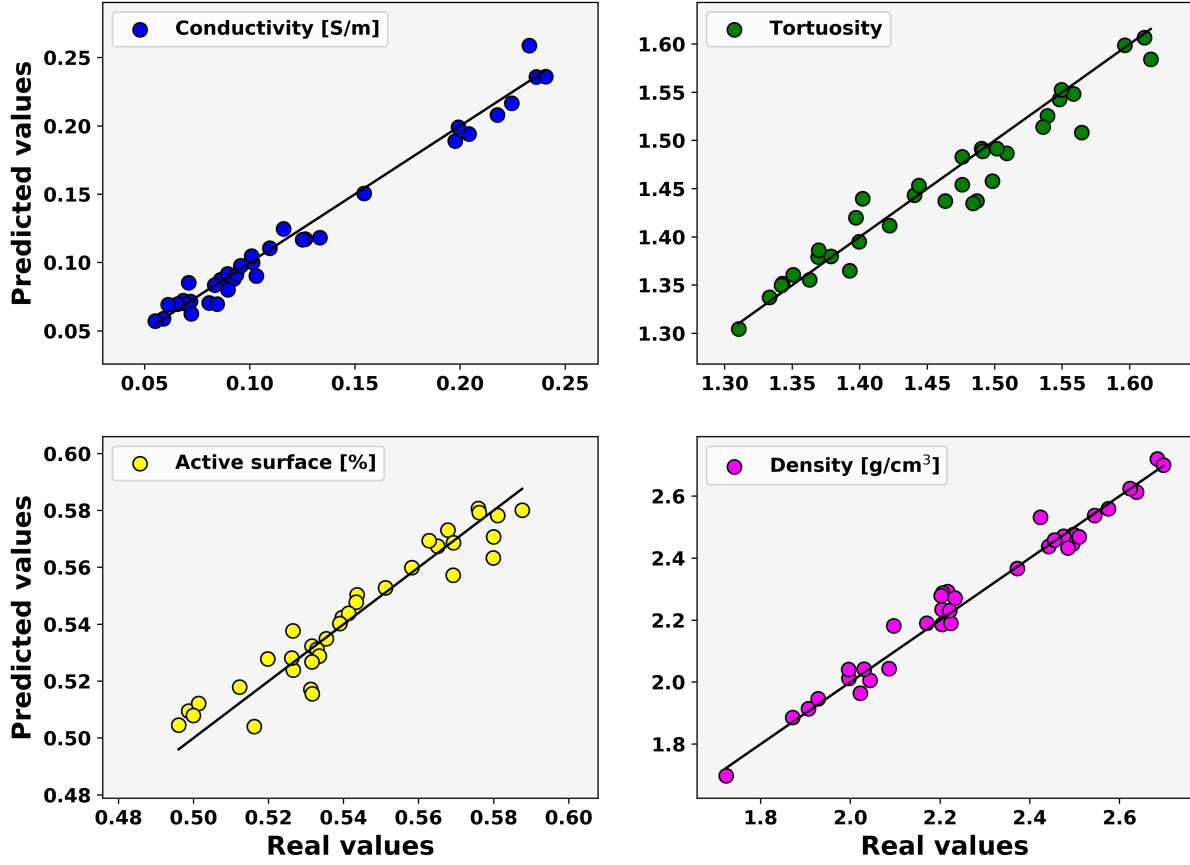


Figure S2 : Regression plots displaying the predictive capabilities of the deterministic learning for each electrode property. This evaluates the goodness of fit by comparing the real values from the testing dataset, versus the values predicted by the deterministic learning. This was done for the case of the electronic conductivity (blue), tortuosity factor (green), active surface area (yellow), and density (magenta).

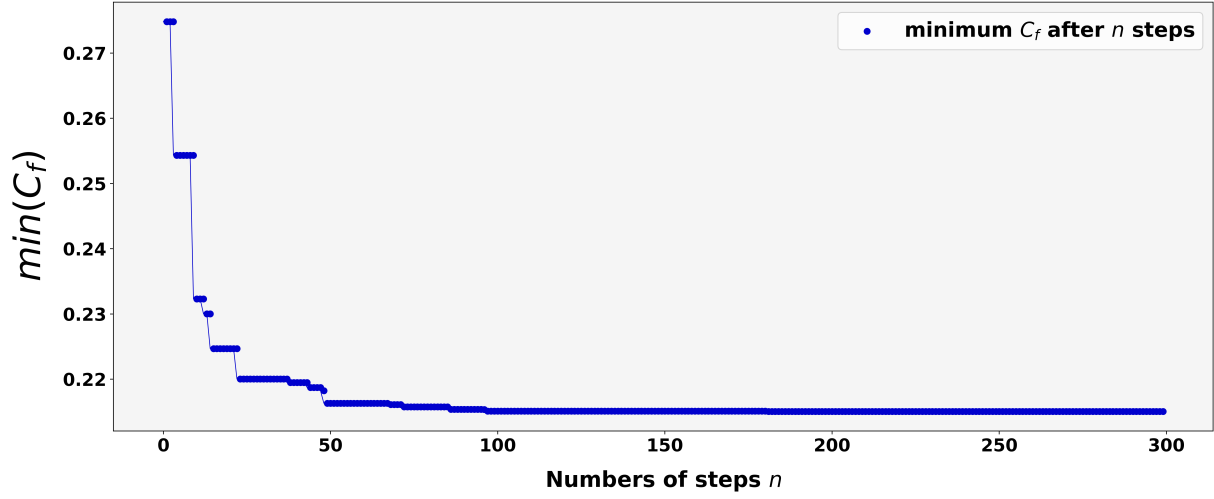


Figure S3 : Convergence plot of the minimum value of the objective function C_f (cf. equation Eq. 3 in the main manuscript) respect to the number of evaluations n within the Bayesian optimization loop. The decrease is very fast within the first steps and starts to plateauing from about step $n^{\circ}100$.

Additional validation metrics

Table S1: Validation metrics calculated over the training dataset, and associated to the fitting of the electrode properties. The table contains the root mean square error in percentage (RMSE%) and the R^2_{score} .

Electrode property	RMSE%	R^2_{score}
Electronic conductivity [S/m]	1.91	0.966
Tortuosity factor	6.88	0.978
Active surface area [%]	1.41	0.909
Density [g/cm ³]	1.87	0.985

Manufacturing parameters for extreme cases

Table S2: Manufacturing parameters corresponding to the four extreme cases that have been compared to the optimal case predicted by the multi-objective optimization in the main manuscript. These four sets of parameters were the inputs of the physics-based models to generate the associated electrodes.

Active material [%]	Solid content [%]	Compression degree [%]	Extreme case
86.3	57.0	33.1	High conductivity case
88.8	56.8	2.24	Low tortuosity factor case
96.5	60.3	9.11	High active surface case
94.5	71.0	36.36	High density case

SISSO algorithm

The SISSO algorithm belongs to the field of regression methods whose main goal is to get a mathematical relationship between a set of input parameters and one output property, by focusing on the non-linear combinations of input parameters for high-dimensional problems. The algorithm lies on two main processes, where one is the feature space construction for the descriptors implementation (function of input parameters), and another one on a solution algorithm for minimizing the prediction error from the model. At the end, the algorithm proposes the following equation (Eq. S1) where the output y is function of descriptors, themselves defined by combinations of input parameters:

$$y = \sum_{i=0}^n c_i \times d_i \quad c_i \neq 0 \quad (\text{Eq. S1})$$

with $(d_i)_{(i \leq n)}$ the descriptors and $(c_i)_{(i \leq n)}$ the coefficients.

- *Feature space:*

The SISSO algorithm is iterative and the definition of descriptors $(d_i)_{(i \leq n)}$ depends on usual algebraic operations to recursively implement them along the algorithm. Indeed, the latter defines an operator set $\hat{H}^{(m)}$ in order to calculate new descriptors using the one calculated at the step before. This means that the complexity of the descriptors increases with the depth of the algorithm,

$$\hat{H}^{(m)} := \{I, +, -, *, | -, ^{-1}, \sqrt{}, ^2, ^3, exp, log\}[\phi_1, \phi_2] \quad (\phi_1, \phi_2) \in \Phi^2 \quad (\text{Eq. S2})$$

with (ϕ_1, ϕ_2) being two elements of the current feature space Φ .

In this algorithm, it is possible to make meaningful combinations regarding the units of the input parameters. Indeed, (m) notifies that dimensional analysis is achieved to retain physical descriptors. In our case study, all the input parameters are dimensionless, simplifying

this concept.

At the first iteration, the feature space contains all available combinations of input parameters using $\hat{H}^{(m)}$. At the second iteration, the feature space contains the latter combinations, including the new combinations of the already calculated ones at the step before. Therefore, the size of the features space grows fast and contains still more complex non-linear combinations of initial input parameters. At the last iteration n of the algorithm, the feature space is defined by:

$$\Phi_n = \bigcup_{i=1}^n H^{(m)}[\phi_1, \phi_2] \quad (\phi_1, \phi_2) \in \Phi_{n-1}^2. \quad (\text{Eq. S3})$$

Regarding the initial problem illustrated by Eq. S1, it is possible to consider the set of descriptors from Φ in order to shift to a matrix problem like $Y = D \times C$, where D is a wide matrix containing all the descriptors.

- *Solution algorithm:*

Such a problem may be solved using common regularized methods for linear problems. However, the size of D requires to narrow the number of descriptors in order to decrease the computational cost for solving the matrix problem. Indeed, SISSO scores all the descriptors with a specific metric (*e.g.* correlation magnitude) and keep the top ranked one for solving the initial problem (called γ). The algorithm focuses on the optimal descriptors and coefficients by minimizing the residual error $\Delta = Y - D_\gamma \times c_\gamma$. Then, it is straightforward to apply l_0 -norm or l_1 -norm regularization.

Tortuosity factor calculations

The tortuosity factor was calculated using the script TauFactor in Matlab.¹ It calculates the tortuosity factor and the effective diffusivity of a porous media, considering the material as mathematically idealized (*i.e.* independent of the physical properties of the fluid). The diffusion in the pores was calculated using the Laplace equation. In this case, an applied concentration gradient was modeled by a concentration difference on the two opposite faces of the electrode microstructure in the direction of the applied concentration gradient. The concentration in every cell in the discretized space of the pores was found by a finite volume method, and the effective diffusivity was obtained by post-processing the gradient field of this concentration. The latter diffusivity of the porous media is a value between 0 and 100 representing the percentage of the diffusivity of the pure fluid. In the other hand, the tortuosity in a particular flow direction is defined as "porosity/effective diffusivity". The experimental tortuosity factors were calculated by using the graphical method for obtaining the total ionic resistance proposed by Landesfeind *et al.* to adjust the impedance curves after assembling symmetric cells with similar cathodes.² The symmetric cell experimental details are described in the Experimental section. An example of the obtained curves is shown in the Figure S4.

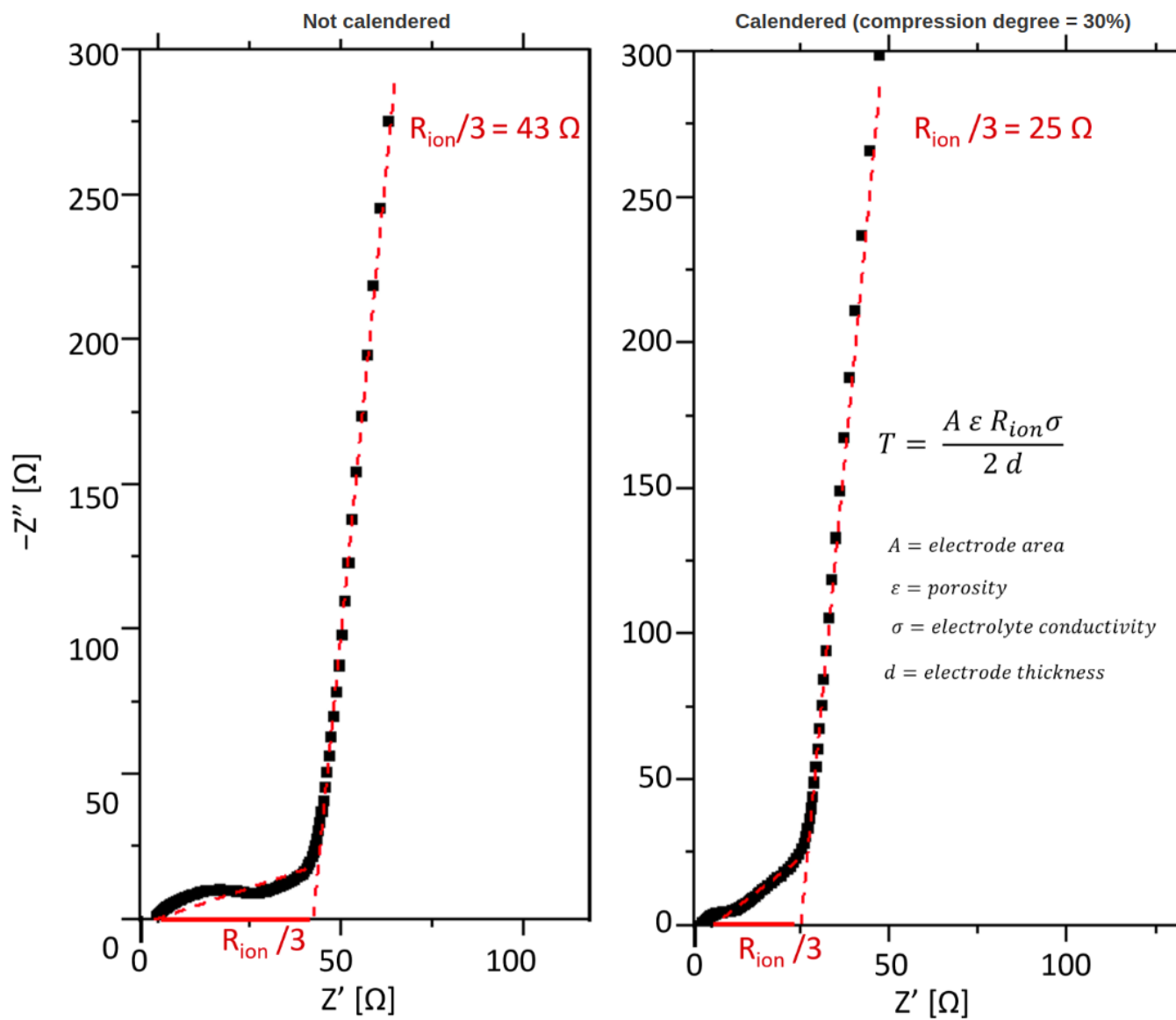


Figure S4 : Characterizations of the tortuosity factors using the impedance spectra.

Comparison of experimental properties

We compared electrode properties coming from different experimental electrodes in order to highlight the results that have been described in the *Experimental* section of the main manuscript. Indeed, we used extreme manufacturing conditions that were included in our previous manuscript Liu *et al.*, Energy Storage Materials, 2023, 54, 156-163 (for physics-based model’s calibration) to support the physical relevance of our theoretical findings arising from the multi-objective optimization.

Table S3: Comparison of electrode properties coming from Liu *et al.*, Energy Storage Materials, 2023, 54, 156-163, with those extracted from the experimental electrode designed with the predicted best manufacturing condition by the multi-objective optimization (last line in the table).³ The experimental conductivity of our prepared electrode was not determined (see the *Experimental* section of the main manuscript).

Active Material [%]	Solid Content [%]	Compression degree [%]	Mass loading [mg/cm ²]	Thickness [μ m]	Density [g/cm ³]	Tortuosity factor	Porosity [%]	Conductivity [S/m]
85	46	0	15.79	121.6	1.55	1.94	59.53	0.029
85	46	10	15.44	108.83	1.73	2.365	55.82	0.0928
85	46	20	15.52	98.5	2.03	2.66	50.91	0.478
90	55	0	15.58	91.67	1.89	1.78	53.03	0.00242
90	55	10	14.94	82.32	2.02	1.78	49.835	0.0432
90	55	20	15.09	72.1	2.33	2.77	42.145	0.14
95	67	0	15.46	77.8	1.09	1.57	51.544	0.000743
95	67	10	15.37	68.2	2.37	1.75	45.0	0.00145
95	67	20	15.11	61.7	2.58	2.47	40.26	0.00164
90	58	30	6.7	48	2.6	1.8	29	N/A

As it can be noticed in the table above, the optimal electrode properties that have been predicted are of high interest. Even though we did not intend to produce the optimal case in the experiment, we demonstrated that preparing an electrode using the results from a multi-objective optimization is meant to obtain consistent electrode properties. In our previous manuscript Liu *et al.*, Energy Storage Materials, 2023, 54, 156-163, we showed that better electrochemical performance was obtained using the formulation of 90% compared to 85% and 95%. The reason is that it can maintain the continuity of the Carbon-Binder Domain (CBD) throughout the electrode, while having a thinner electrode and lower interfacial resistance between the electrode and the current collector under the same calendering degree.³ This highlights the power of our proposed approach detailed in the main manuscript.

References

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